```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
from google.colab import files

# Upload image
uploaded = files.upload()

# Get uploaded image name
image_path = list(uploaded.keys())[0]

# Load image
img = cv2.imread(image_path)
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving imagesof cycle.jpg to imagesof cycle.jpg

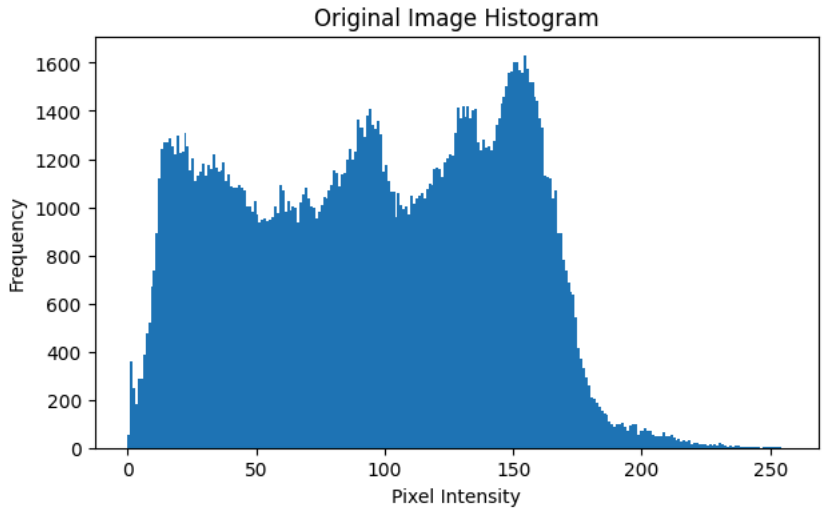
```
# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Display original grayscale image
plt.figure(figsize=(6, 5))
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```




```
# 1. Original Histogram

plt.figure(figsize=(7, 4))
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Original Image Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.show()
```




```
# 2. Contrast Stretching

min_val = np.min(gray)
max_val = np.max(gray)

stretched = (
    (gray - min_val) *
    (255.0 / (max_val - min_val))
).astype(np.uint8)

plt.figure(figsize=(6, 5))
plt.imshow(stretched, cmap="gray")
plt.title("Contrast Stretched Image")
plt.axis("off")
plt.show()
```




```
# 3. Histogram Equalization

equalized = cv2.equalizeHist(gray)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(equalized, cmap="gray")
plt.title("Histogram Equalized Image")
```

```
plt.axis("off")

plt.show()
```






```
# 4. CLAHE

clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

clahe_image = clahe.apply(gray)

plt.figure(figsize=(6, 5))
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE Enhanced Image")
plt.axis("off")
plt.show()
```




```
# 5. Compare Histograms

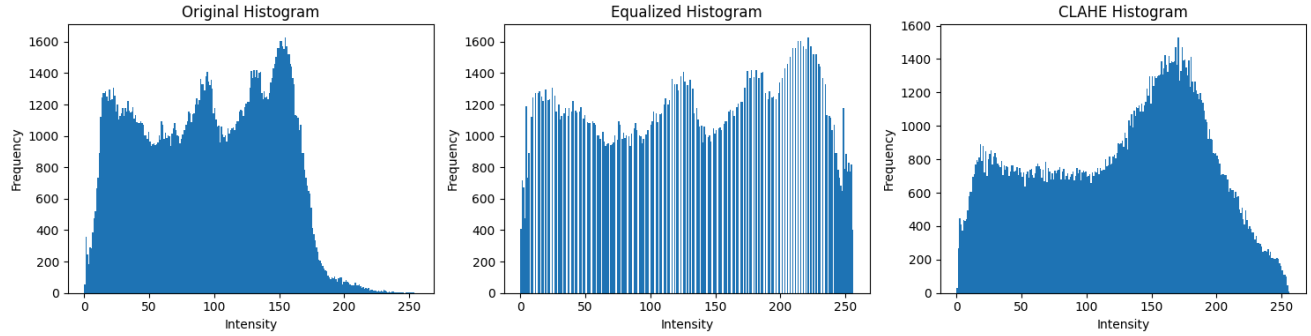
plt.figure(figsize=(15, 4))

plt.subplot(1, 3, 1)
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Original Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")
```

```
plt.subplot(1, 3, 2)
plt.hist(equalized.ravel(), bins=256, range=[0, 256])
plt.title("Equalized Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.subplot(1, 3, 3)
plt.hist(clahe_image.ravel(), bins=256, range=[0, 256])
plt.title("CLAHE Histogram")
plt.xlabel("Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```




```
# 6. Final Image Comparison

plt.figure(figsize=(16, 5))

plt.subplot(1, 4, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1, 4, 2)
plt.imshow(stretched, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(1, 4, 3)
plt.imshow(equalized, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(1, 4, 4)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

plt.tight_layout()
plt.show()
```


Original
Contrast Stretching
Histogram Equalization
CLAHE